

SS1 Science Track: Chemistry Diagnostic (Part 1)

Standard: Cambridge IGCSE / O-Level Bridge

Cluster 1: Particles, Atomic Structure & Bonding (Questions 1 - 20)

Note: Assume Standard Temperature and Pressure (STP) where molar volume = 22.4 dm^3 . Relative Atomic Masses (A_r): H=1, C=12, O=16, S=32, Cl=35.5.

No.	Question & Options	Ans	Diagnostic Focus
1	Which state of matter consists of particles that are far apart and move randomly at high speeds? A) Solid B) Liquid C) Gas D) Plasma	C	Kinetic Theory (Particle behavior).
2	The process by which a solid changes directly into a gas without melting is called: A) Evaporation B) Sublimation C) Condensation D) Diffusion	B	Changes of State.
3	Bromine vapor spreads through a glass jar. Which process explains this? A) Filtration B) Crystallization C) Volatilization D) Diffusion	D	Kinetic Theory (Diffusion).
4	An atom of an element has an atomic number of 11 and a mass number of 23. How many neutrons does it have? A) 11 B) 23 C) 12 D) 34	C	Atomic Structure (Neutron calculation).
5	Isotopes of the same element have different numbers of _____. A) Protons B) Electrons C) Neutrons D) Valence electrons	C	Definition of Isotopes.
6	What is the electronic configuration of a Chlorine atom ($Z=17$)? A) 2, 8, 8 B) 2, 10, 5 C) 2, 8, 7 D) 8, 8, 1	C	Electron Shell arrangement.
7	Ionic bonding usually occurs between: A) Two metals B) A metal and a non-metal C) Two non-metals D) Two noble gases	B	Types of Chemical Bonding.
8	Which of the following substances has a giant covalent structure? A) Sodium Chloride B) Water C) Diamond D) Methane	C	Bonding Models & Properties.
9	What happens to a Magnesium atom when it forms a Mg^{2+} ion? A) It gains 2 electrons B) It loses 2 electrons C) It gains 2 protons D) It loses 2 protons	B	Ion formation & Oxidation.
10	The formula for Aluminum Sulfate is: A) AlSO_4 B) Al_2SO_4 C) $\text{Al}_3(\text{SO}_4)_2$ D) $\text{Al}_2(\text{SO}_4)_3$	D	Chemical Formula Writing (Valency).
11	Calculate the Relative Molecular Mass (M_r) of Sulfuric Acid, H_2SO_4 . A) 49 B) 98 C) 100 D) 150	B	Quantitative Chemistry (M_r calculation).
12	How many moles are in 88g of Carbon Dioxide (CO_2)? A) 1 mole B) 2 moles C) 0.5 moles D) 4 moles	B	The Mole Concept (Mass to Moles).

No.	Question & Options	Ans	Diagnostic Focus
13	A separation technique used to obtain pure water from a salt solution is: A) Filtration B) Simple Distillation C) Chromatography D) Fractionating column	B	Separation Techniques (Purification).
14	Which of these moves fastest during paper chromatography? A) The solvent B) The most soluble dye C) The least soluble dye D) The baseline	B	Chromatography Theory (R_f factors).
15	Calculate the percentage by mass of Oxygen in MgO . ($Mg=24$, $O=16$) A) 40% B) 60% C) 16% D) 24%	A	Percent Composition.
16	Element X is in Group II and Period 3 of the Periodic Table. X is: A) Aluminum B) Magnesium C) Calcium D) Boron	B	Periodic Table Navigation.
17	When an element in Group VII reacts, it becomes an ion with a charge of: A) +1 B) -1 C) +7 D) -7	B	Group Trends and Valency.
18	Noble gases are unreactive because they: A) Are gases B) Have a full outer shell of electrons C) Are at the end of the table D) Have no neutrons	B	Stability and Reactivity.
19	Which type of bond involves the "delocalized sea of electrons"? A) Ionic B) Covalent C) Metallic D) Hydrogen	C	Metallic Bonding.
20	Diagnostic Insight: If a student fails Q10 and Q12, they lack the foundational stoichiometry required for SS1 Chemistry and will struggle in the Science track.	-	Quantitative Foundation Check.